

TYPES OF PROMPTS: ZERO-SHOT, ONE-SHOT, AND FEW-SHOT PROMPTING

1. Introduction

Generative AI models are designed to understand natural language instructions and generate responses based on the information provided in a prompt. However, an AI model does not always need the same amount of guidance for every task. For some tasks, a simple instruction is sufficient, while other tasks may require examples to demonstrate exactly how the expected output should look. Based on the number of examples provided in a prompt, prompting can be broadly classified into **zero-shot prompting, one-shot prompting, and few-shot prompting**.

These prompting techniques are important because they provide different levels of guidance to a generative AI model. **Zero-shot prompting** asks the model to perform a task without providing any example, **one-shot prompting** provides one example, and **few-shot prompting** provides multiple examples. Understanding these techniques helps users select an appropriate prompting strategy depending on the complexity of the task, the desired output, and the behavior they want from the AI model.

2. What are Types of Prompts?

A **type of prompt** refers to a particular method of designing a prompt based on the amount of guidance or examples provided to a generative AI model.

The three fundamental example-based prompting approaches are:

1. **Zero-Shot Prompting**
2. **One-Shot Prompting**
3. **Few-Shot Prompting**

The primary difference between these approaches is the **number of examples included in the prompt**.

Prompt Type	Number of Examples	Level of Guidance
Zero-shot	0	Low
One-shot	1	Moderate
Few-shot	2 or more	High

These approaches can be used for tasks such as classification, translation, text generation, information extraction, question answering, summarization, and sentiment analysis.

3. Zero-Shot Prompting

3.1 Definition

Zero-shot prompting is a prompting technique in which the AI model is asked to perform a task **without being provided with any examples of the expected input-output behavior**.

The user directly describes the task through natural language instructions, and the model attempts to perform the task using the knowledge and patterns it learned during pre-training. The term *zero-shot* means that the model receives **zero examples** for the specific task within the prompt.

Example

Consider the following prompt:

Prompt: Classify the following sentence as Positive, Negative, or Neutral:

"The camera quality is excellent."

The model has not been given any examples showing how positive, negative, or neutral sentences should be classified. It must understand the instruction and determine the appropriate category.

Expected Output:

Positive

3.2 Structure of Zero-Shot Prompting

A basic zero-shot prompt generally contains:

Instruction → Input → Expected Task

For example:

Instruction: Translate the following sentence into Telugu.

Input: "Artificial intelligence is changing the world."

Task: Provide only the Telugu translation.

No example is provided to the model.

3.3 Working of Zero-Shot Prompting

The user first defines the task clearly and then provides the input that needs to be processed. The AI model interprets the instruction and uses its previously learned knowledge to determine how the task should be performed.

For example:

"Summarize the following paragraph in three sentences."

The model does not need to be shown an example of a summary. It understands the meaning of *summarize* from its training and generates a three-sentence summary.

The effectiveness of zero-shot prompting therefore depends heavily on the model's existing capabilities and the clarity of the instruction.

3.4 Examples of Zero-Shot Prompting

Example 1: Translation

Translate the following sentence into Hindi:

"Machine learning is a subset of artificial intelligence."

Example 2: Classification

Classify the following review as Positive, Negative, or Neutral:

"The laptop has excellent performance."

Example 3: Summarization

Summarize the following paragraph in two sentences.

Example 4: Question Answering

What are the major applications of generative AI?

In all these examples, the model receives **no demonstration examples**.

3.5 Advantages of Zero-Shot Prompting

- **Simple to design** because no examples are required.
 - **Consumes fewer tokens** because the prompt does not contain demonstrations.
 - **Reduces prompt length** and can therefore reduce computational cost.
 - **Works well for common and well-defined tasks** that the model already understands.
 - **Easy to modify** when the task requirements change.
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3.6 Limitations of Zero-Shot Prompting

- May produce inconsistent results for complex tasks.
 - The model may interpret ambiguous instructions incorrectly.
 - It provides limited guidance about the expected output format.
 - Performance can decrease when the task is specialized or unusual.
 - The model may make assumptions that differ from the user's expectations.
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4. One-Shot Prompting

4.1 Definition

One-shot prompting is a prompting technique in which **one example** of the desired input-output behavior is provided before asking the model to perform the task on a new input.

The example demonstrates the expected relationship between an input and its corresponding output. The model uses this demonstration as a pattern and applies the learned pattern to the new input.

The important characteristic of one-shot prompting is that exactly **one demonstration example** is provided.

4.2 Example of One-Shot Prompting

Suppose we want the model to classify movie reviews.

Example:

Review: "The movie was fantastic and entertaining."

Sentiment: Positive

Task:

Review: "The movie was boring and poorly written."

Sentiment:

Expected Output:

Negative

The first review provides the model with an example of how the classification task should be performed.

4.3 Structure of One-Shot Prompting

A one-shot prompt generally follows this structure:

Example → New Input → Expected Output

For example:

Example:

English: "Good morning."

Telugu: "శుభోదయం"

Input:

English: "Good night."

Telugu:

The model uses the first example to understand the expected translation pattern.

4.4 Working of One-Shot Prompting

The model first analyzes the provided example and identifies the relationship between the input and output. It then receives a new input and attempts to apply the same pattern.

For example:

Example:

$2 + 2 = 4$

Question:

5 + 3 = ?

The example establishes the expected response pattern. The model then applies the same type of reasoning to the new question.

One-shot prompting is particularly useful when the task is easy to describe but the **expected output style or format needs clarification**.

4.5 Examples of One-Shot Prompting

Example 1: Sentiment Classification

Review: "The product is amazing."

Sentiment: Positive

Review: "The product broke within two days."

Sentiment:

Output:

Negative

Example 2: Text Classification

Text: "The football match was exciting."

Category: Sports

Text: "The new processor consumes less power."

Category:

Output:

Technology

Example 3: Translation

English: "How are you?"

Telugu: "మీరు ఎలా ఉన్నారు?"

English: "What is your name?"

Telugu:

The example demonstrates the expected translation format.

4.6 Advantages of One-Shot Prompting

- Provides more guidance than zero-shot prompting.
- Helps clarify the expected output format.

- Can improve performance on tasks with specific patterns.
 - Requires fewer tokens than few-shot prompting.
 - Useful when one carefully selected example represents the desired behavior.
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4.7 Limitations of One-Shot Prompting

- A single example may not represent all possible cases.
 - A poorly selected example can confuse the model.
 - It may not be sufficient for complex classification tasks.
 - The model may overgeneralize from the single example.
 - Performance can depend heavily on the quality of the demonstration.
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5. Few-Shot Prompting

5.1 Definition

Few-shot prompting is a prompting technique in which **multiple examples** are provided to the AI model before asking it to perform the task on a new input.

These examples demonstrate different input-output patterns and provide the model with a clearer understanding of the expected task behavior.

The number of examples can vary depending on the task, model, available context window, and token budget. For example, a prompt may contain three, five, ten, or more examples.

5.2 Example of Few-Shot Prompting

Consider sentiment classification:

Example 1:

Review: "The product is excellent."

Sentiment: Positive

Example 2:

Review: "The product is completely useless."

Sentiment: Negative

Example 3:

Review: "The product arrived yesterday."

Sentiment: Neutral

New Review:

"The product works perfectly and I am very satisfied."

Sentiment:

Expected Output:

Positive

Here, three examples are provided before the actual input, making this a **few-shot prompt**.

6. How Few-Shot Prompting Works

Few-shot prompting works by providing the model with several demonstrations of the desired behavior.

Consider the following structure:

Input 1 → Output 1

Input 2 → Output 2

Input 3 → Output 3

New Input → ?

The model examines the examples and identifies the common pattern connecting the inputs and outputs. It then applies the learned pattern to the new input.

For example:

Text: "Apple released a new iPhone."

Category: Technology

Text: "India won the cricket match."

Category: Sports

Text: "The restaurant serves delicious food."

Category: Food

Text: "The processor provides higher performance."

Category:

The examples help the model infer that the expected output is a category label.

Expected Output:

Technology

7. Types of Examples Used in Few-Shot Prompting

The examples selected for few-shot prompting can serve different purposes.

7.1 Format Examples

Examples can demonstrate the exact structure in which the model should generate its response.

Input: Python

Output: Programming Language

7.2 Classification Examples

Examples can demonstrate how different inputs should be assigned to different categories.

"I loved the movie." → Positive

"I hated the movie." → Negative

"The movie was released yesterday." → Neutral

7.3 Style Examples

Examples can demonstrate the writing style expected from the model.

For example, examples can show whether the output should be formal, academic, conversational, humorous, technical, or simplified.

7.4 Transformation Examples

Examples can demonstrate how one form of information should be transformed into another.

For example:

Sentence → Summary

English → Telugu

Question → SQL Query

Natural Language → JSON

8. Advantages of Few-Shot Prompting

- Provides stronger guidance than zero-shot and one-shot prompting.
 - Helps the model understand complex or specialized tasks.
 - Demonstrates the desired output format clearly.
 - Can improve classification accuracy for custom categories.
 - Allows users to define task-specific behavior without changing the model's parameters.
 - Helps handle tasks where instructions alone are insufficient.
 - Can demonstrate multiple possible input-output patterns.
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9. Limitations of Few-Shot Prompting

- Requires more tokens because multiple examples are included.
- Longer prompts can increase computational cost.
- Poor-quality examples can negatively influence the generated output.
- Too many examples may consume a large portion of the model's context window.

- Selecting representative examples can require significant effort.
- Examples can introduce unintended biases into the model's responses.

10. Comparison of Zero-Shot, One-Shot, and Few-Shot Prompting

Feature	Zero-Shot	One-Shot	Few-Shot
Examples	0	1	2 or more
Guidance	Low	Moderate	High
Prompt size	Small	Medium	Larger
Token usage	Low	Moderate	Higher
Implementation	Very easy	Easy	More careful design required
Best suited for	Simple, familiar tasks	Tasks requiring one pattern	Complex or specialized tasks
Output pattern demonstration	No	One example	Multiple examples
Dependency on examples	None	High	High
Flexibility	High	Moderate	High
Cost	Generally lower	Moderate	Generally higher

11. Same Task Using All Three Prompting Techniques

To clearly understand the difference, consider a **sentiment classification task**.

Zero-Shot

Classify the following review as Positive or Negative:

"The product quality is excellent."

Output:

Positive

There are **zero examples**.

One-Shot

Review: "The product quality is terrible."

Sentiment: Negative

Review: "The product quality is excellent."

Sentiment:

Output:

Positive

There is **one example**.

Few-Shot

Review: "The product quality is terrible."

Sentiment: Negative

Review: "The product is amazing and reliable."

Sentiment: Positive

Review: "The product arrived yesterday."

Sentiment: Neutral

Review: "The product quality is excellent."

Sentiment:

Output:

Positive

There are **multiple examples**.

The task remains the same in all three cases, but the amount of guidance provided to the model changes.

12. When to Use Each Prompting Technique?

Use Zero-Shot Prompting When:

- The task is simple and clearly understood by the model.
- No specific examples are required.
- Prompt length and token usage should be minimized.
- The desired output can be clearly described through instructions.

Use One-Shot Prompting When:

- One example is enough to demonstrate the expected behavior.
- The output format needs clarification.
- The task has a simple but specific pattern.
- You want additional guidance without significantly increasing prompt length.

Use Few-Shot Prompting When:

- The task is complex or specialized.
 - Multiple examples are necessary to explain the expected behavior.
 - The model needs to distinguish between several categories.
 - The desired output pattern is difficult to explain using instructions alone.
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13. Key Differences at a Glance

The fundamental difference between these three prompting techniques can be remembered using the following rule:

Zero-Shot → No Example

One-Shot → One Example

Few-Shot → Multiple Examples

In other words, the word "**shot**" refers to the number of examples or demonstrations provided to the model within the prompt.

14. Important Considerations for Designing Examples

When using one-shot or few-shot prompting, the quality of the examples is extremely important.

14.1 Choose Relevant Examples

Examples should be closely related to the actual task that the model needs to perform.

14.2 Use Clear Examples

Examples should be easy for the model to understand and should not contain unnecessary information.

14.3 Maintain Consistent Formatting

All demonstrations should follow a consistent input-output structure.

14.4 Include Representative Cases

For classification tasks, examples should represent the different categories that the model may encounter.

14.5 Avoid Contradictory Examples

Examples that contain inconsistent or conflicting labels can cause the model to learn an incorrect pattern.

15. Real-World Applications

Zero-shot, one-shot, and few-shot prompting can be applied across many domains.

Domain	Example Application

Education	Generate explanations according to student level
Healthcare	Classify medical text into predefined categories
Software Development	Generate code from natural-language requirements
Customer Support	Classify customer queries
Finance	Categorize financial documents
Marketing	Generate content in a specified style
NLP	Perform sentiment and intent classification
Translation	Translate text according to demonstrated patterns
Data Processing	Extract structured information from documents
Research	Summarize and classify research papers

16. Chapter Summary

Zero-shot, one-shot, and few-shot prompting are fundamental prompting techniques used to control how much task-specific guidance is provided to a generative AI model.

Zero-shot prompting provides instructions without examples and is suitable when the task is straightforward and the model already understands the required behavior. **One-shot prompting** provides a single example to demonstrate the desired pattern, making it useful when one demonstration can clarify the expected output. **Few-shot prompting** provides multiple examples and is particularly useful for complex, specialized, or pattern-based tasks.

The three techniques can therefore be viewed as increasing levels of demonstration:

Zero-Shot → Instruction Only

One-Shot → Instruction + One Example

Few-Shot → Instruction + Multiple Examples

The selection of an appropriate prompting technique depends on the **complexity of the task, model capability, quality of available examples, desired output, context-window limitations, and token budget**.